

Dexter Fernandes

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Machine Learning Engineer

Machine learning engineer with 3.5 years experience collaborating with cross-functional teams, to ship computer vision solutions on the edge for clients that include British Petroleum and Austin Industries. I own the full-stack: handle large datasets with DVC, build models with PyTorch, track experiments with MLFlow and deploy via Azure. As of June 2025, my systems monitor safety and productivity on construction sites in Texas and on-shore oil rigs in Oman, boosting detection accuracy by 15%.

WORK EXPERIENCE

Esbaaar • Muscat, Oman

01/2023 - 08/2023

AI Engineer

- Spearheaded the end-to-end development and production rollout of YOLOv5 models, to elevate rig worker and equipment detection precision by 15% in 3 months.
- Developed time-series machine learning models for the classification of on-shore oil rig operations to increase worker safety and reduce supervision required during 24/7 operation, achieving a 23% higher F1 score than human experts.
- Built an automated sensor-data cleansing pipeline (3.6 M data points) that cut model roll out from 3 days to less than 24 hours.
- Led strategic MLOps initiatives to continuously enhance experimentation and deployment workflows through bayesian hyperparameter tuning of time-series and vision models.
- Refactored a legacy monolithic system into a Kafka-driven microservice architecture, containerising each service with Docker to enhance failure isolation and improve MTBF by 35% within four months
- Designed a dynamic memory management solution to address recurring crashes in a live video streaming system, reducing crash frequency by 67%.
- Enhanced code quality and documentation standards by leading formal peer code reviews and quality-assurance activities within each sprint cycle, ensuring best practices.

LivNSense Technologies Pvt. Ltd. • Bengaluru, India

05/2020 - 12/2022

Lead Software Engineer - AI • Full-time

- Led a 4-engineer team on ViCAS to design, train, and deploy deep learning models for real-time object detection, pose estimation, and tracking on edge hardware to reduce false positives by 33% while achieving sub-50 ms inference latency within 6 months.
- Implemented real-time object tracking with YOLO-DeepSORT, maintaining consistent identities across frames and enabling reliable downstream analytics.
- Repurposed legacy object detection models for automated labelling pipeline workflows, reducing manual annotation effort by 80% and accelerating dataset curation.
- Engineered synthetic training data pipelines using OpenCV and NumPy, reducing edge-case model development time from 6 days to 2 days for high-risk production environments.
- Developed core perception algorithms on resource-constrained Intel edge devices, leveraging mixed-precision and fixed-point quantization with structured pruning to meet strict memory and latency budgets for real-time decision-making in industrial environments.
- Optimized neural network architectures using quantization-aware training, structured pruning, and layer fusion thus reducing model size by 50%.
- Implemented rigorous testing and validation frameworks based on simulated adverse conditions, to ensure model accuracy and stability, reducing performance variance by 25% under challenging lighting, occlusion, and noise scenarios.

EDUCATION

Master of Science in Robotics - Merit

University of Bristol • Bristol, England

09/2023 - 09/2024

Bachelor of Technology in Electrical and Electronics Engineering

Visvesvaraya National Institute of Technology • Nagpur, India

07/2016 - 05/2020

CERTIFICATIONS

Deep Learning Specialisation

deeplearning.ai

07/2019

SKILLS

- **Perception:** LiDAR SLAM, Nav2, Neural Network Compression Framework (NNCF), OpenCV, OpenVINO, PyTorch, ROS2, Sci-kit Learn, TensorFlow, Vision Transformer (ViT), YOLOv5/v8
- **Backend & Cloud / Edge:** Apache Kafka, AWS, FastAPI, Intel VPU, Microsoft Azure, NVIDIA Jetson AGX, PostgreSQL
- **DevOps & MLOps:** Docker, DVC (Data Version Control), Git, Linux, MLFlow, Optuna